

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1– Concept Mapping

Course Code:	CSA0312	Course Title:	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 1 – Concept Mapping
Weightage:	10% of CIA	Total Marks:	100
CO1 Statement:	Basics, Data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C', Primitive and non-primitive data structure, Linear and non-linear data structure Arrays & Recursion, Performance analysis and Measurement: Time and space analysis of algorithms, Average, best and worst-case analysis.		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Code of Conduct

I (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

Instructions

- This test contains 2 concept mapping, Total: 100 Marks.
- Evaluation of Answer Quality -Checking whether the answer includes: Concept Identification , Linkages, Organization, Integration of Literature, and Clarity
- No DTP printouts or electronic devices permitted. They should provide materials in PDF format.

Q4 A ride-sharing application must efficiently match users with nearby drivers in real time. Construct a concept map connecting Algorithm → Input Size → Time Complexity → Space Complexity → Efficiency → Performance Measurement. Illustrate how increasing input size impacts system performance and establish relationships between these concepts. Analyze how the choice of algorithm affects efficiency and justify the most suitable approach.

